

CSSE290: Artificial Life

Final Project/Presentation

Final Class before Spring Break

Artificial Life Simulator

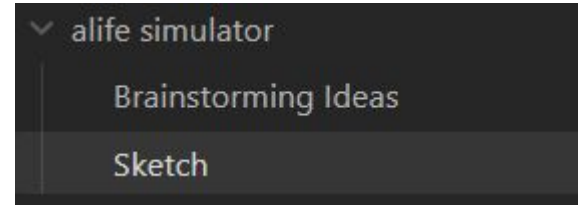
Week 5 - ~~you should have selected one and have it approved by instructor~~ You should start exploring the list of artificial life simulators or doing your own search. You may wish to use some of your time over the break to look these without the stress of other deadlines

Week 10 - you will give a presentation on it

You should do the following:

Learn about it and answer questions such as:

- What is the simulator focused on?
- What levels does it attempt to capture?
- What is an example of a research question it purports to be able to help answer?



There are some “games” that may still be perfectly qualified to answer questions. It will be up to you to demonstrate what kind of questions it can answer.

Artificial Life Simulator

Prepare a presentation (~10 ~25 minutes) that:

- Introduces the simulator
- Explains the components
- Gives an overview of adjustable parameters
- Explains a specific experiment that has been conducted using it

-----NEW-----

Report on *INTERESTING/SURPRISING/INSIGHTFUL* results of performing an experiment using it.

I expect you will have to run many experiments to find an interesting one.

You can focus on the most interesting experiment, you don't have to report on all of them.

Artificial Life Simulator - NEW

Where applicable it will be best to provide additional analysis/investigation into the results:

- Is it repeatable?
- What are the most important mechanisms (change/remove components)?
- Can we understand in detail one specific instance of the observed results/behavior?
- Could we possibly understand the system better through information theory?
 - How could we apply IT to study this?
- Could we possibly understand the system better through dynamical systems theory?
 - How could we apply DST to study this?
- Is there potential of seeing this behavior/phenomena in the real world?
 - How could you try to imitate this?
 - What would be crucial to consider in evaluating this?

Alex Brickley - AVIDA

Artificial life platform for research

Population

Avida-ED File Freezer Control Help

Population Organism Analysis

Freezer

- Configured Dishes
 - @default
- Organisms
 - @all_functions
 - @ancestor
- Populated Dishes
 - @example

Workspace: 'Default'

- Save workspace if you want to use items saved to the freezer after this session

Time: (not started)

Mode: Fitness

Freeze New Run Forward

Selected Organism Type

Name:
Fitness:
Energy Acq. Rate:
Offspring Cost:
Age (updates):
Ancestor:
Viable:

Population Statistics

Size (# of organisms):
Avg. Fitness:
Avg. Energy Acq. Rate:
Avg. Offspring Cost:
Avg. Age (updates):
Ancestor(s) at start:
of viable organisms:

Function	Performed	Function	# of Organisms
NOT		NOT	
NAN		NAN	
AND		AND	
ORN		ORN	
ORO		ORO	
ANT		ANT	
NOR		NOR	
XOR		XOR	
EQU		EQU	

Resource	Not	Nan	And	Orn	Oro	Ant	Nor	Xor	Equ
Type									
Amount									

Lft Y-axis (Ave. Fitness) Rt Y-axis (None)

Pause Run at Update 1000 when checked

Organism Specific

<<

Avida-ED

File

Freezer

Control

Help

Setup

Stats

>>

 Population

 Organism

 Analysis

Freezer

▼ Configured Dishes

@default

▼ Organisms

@all_functions

@ancestor

▼ Populated Dishes

@example

 Workspace: 'Default'

- Save workspace if you want to use items saved to the freezer after this session

Function	Times Performed
0 NOT	0
1 NAN	0
2 AND	0
3 ORN	0
4 ORO	0
5 ANT	0
6 NOR	0
7 XOR	0
8 EQU	0

Input Buffer

Registers: AX, BX, CX

Stack A (first two frames)

Stack B (first two frames)

Last Output

Instruction Details

► Just Executed:

► About to Execute:

Cycle: 0

Reset Back Run Forward End

Analysis Mode

The screenshot shows the Avida-ED software interface in Analysis Mode. The window has a menu bar with '<<', 'Avida-ED', 'File', 'Freezer', 'Control', and 'Help'. On the left is a sidebar with three icons (a 4x4 grid, a circular arrow, and a line graph) and buttons for 'Population', 'Organism', and 'Analysis'. Below these is a 'Freezer' section with expandable categories: 'Configured Dishes' (containing '@default'), 'Organisms' (containing '@all_functions' and '@ancestor'), and 'Populated Dishes' (containing '@example'). At the bottom left of the sidebar, a trash can icon is next to the text 'Workspace: Default' and a note: '- Save workspace if you want to use items saved to the freezer after this session'. The main area is a large, empty light gray rectangle. The bottom of the window contains a control panel with three rows of color selection (Red, Blue, Black), each with a dropdown arrow. To the right of these are two labels with dropdown arrows: 'Average Fitness' (labeled 'Thicker line; left y-axis') and 'Average Offspring Cost' (labeled 'Thinner line; right y-axis').

<< Avida-ED File Freezer Control Help

Population
Organism
Analysis

Freezer

- ▼ Configured Dishes
 - @default
- ▼ Organisms
 - @all_functions
 - @ancestor
- ▼ Populated Dishes
 - @example

Workspace: Default
- Save workspace if you want to use items saved to the freezer after this session

X Red
X Blue
X Black

Average Fitness Thicker line; left y-axis
Average Offspring Cost Thinner line; right y-axis

Freezer



Stores experimental setups with specific settings, for replication

Stores individuals for future use or analysis.

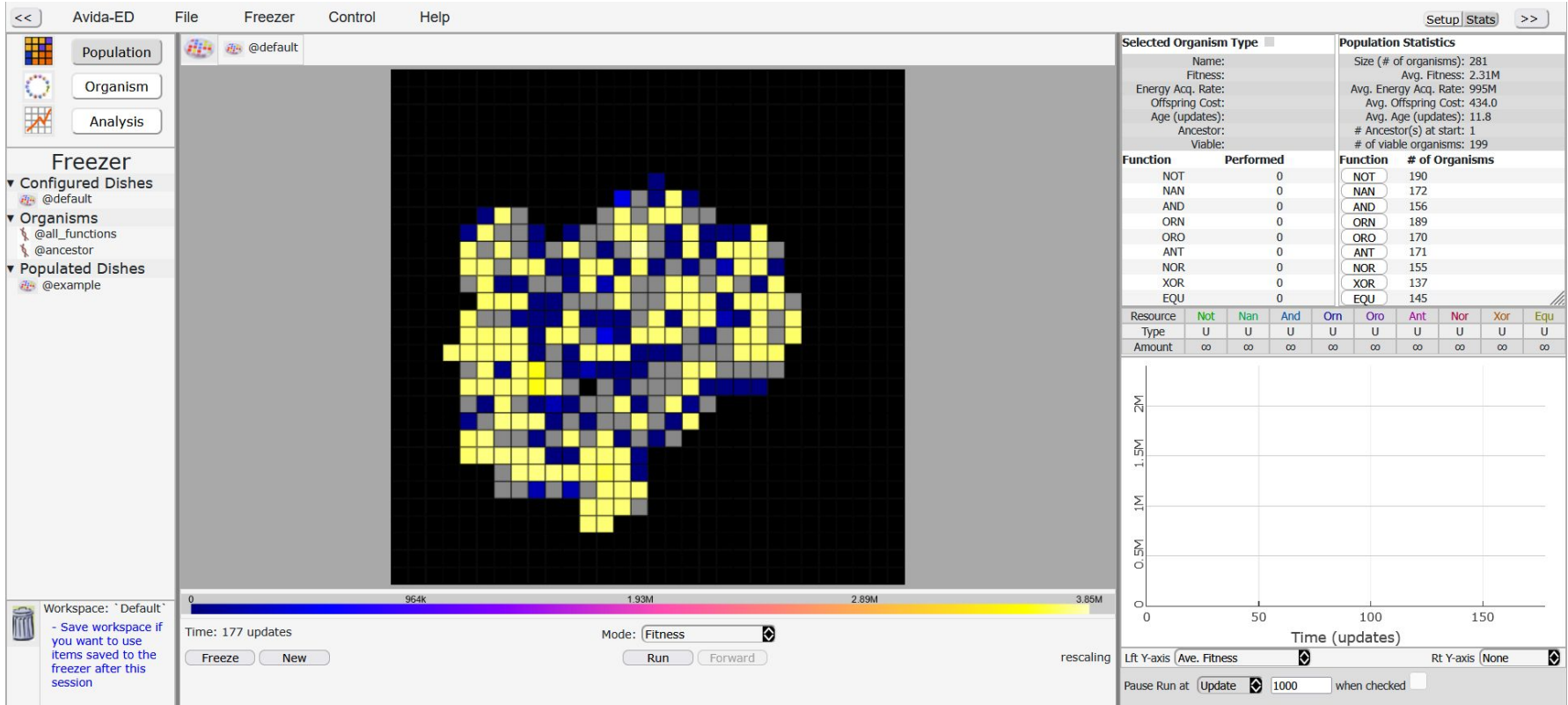
Stores populations for future use or analysis.

Example population

The screenshot displays the Avida-ED web interface in a browser window. The address bar shows the URL `avida-ed.msu.edu/app4/index.html`. The interface is divided into several sections:

- Top Bar:** Contains navigation links like "Population", "Organism", and "Analysis".
- Left Panel:** Features a "Freezer" section with a list of "Configured Dishes" and "Organisms". The "Example" dish is selected.
- Main Area:** A large grid representing the simulation environment, currently showing a dark, mostly empty space.
- Right Panel:** Contains "Environmental Settings" and "Resource Options". The "Environmental Settings" section includes fields for "Dish Size" (30 x 30 cells), "Per Site Mutation Rate" (2.0 %), and "Ancestral Organism(s)". The "Resource Options" section lists various resources (e.g., Nitro, Nitro, Nitro, Nitro, Nitro, Nitro, Nitro, Nitro, Nitro, Nitro) with their respective "Unlimited" status and "Easy" or "Hard" difficulty levels.
- Bottom Bar:** Includes a "Workspace" section with a "Save workspace" button, a "Time" display (0), a "Mode" dropdown (Fitness), and buttons for "Run" and "Forward".

Example population



Example of genome replication

The screenshot displays the Avida-ED software interface, which is used for simulating digital evolution. The interface is divided into several panels:

- Top Panel:** Contains browser tabs and the address bar showing `avida-ed.msu.edu/app-4/index.html`.
- Left Panel:** Includes buttons for **Population**, **Organism**, and **Analysis**. Below these are sections for **Freezer** (Configured Dishes, Organisms, Populated Dishes) and a **Workspace** note.
- Center Panel:** Shows a circular genome visualization with a sequence of colored dots representing different functions. The sequence is labeled `@all_functions`.
- Right Panel:** Contains a table of function counts and a section for input buffers and registers.

Function	Times Performed
0 NOT	0
1 NAND	0
2 AND	0
3 ORN	0
4 ORO	0
5 ANT	0
6 NOR	0
7 XOR	0
8 EQU	0

Input Buffer: [Visual representation of input buffer]

Registers: AX, BX, CX [Visual representation of registers]

Stack A (first two frames) [Visual representation of stack A]

Stack B (first two frames) [Visual representation of stack B]

Last Output: [Visual representation of last output]

Instruction Details:

- Just Executed:
- About to Execute:

At the bottom, there is a progress bar with markers at 0, 72, 143, 215, 287, 358, and 430. Below the progress bar, it says "Cycle: 0" and "Reset Back Run Forward End".

Does one not



Reset Back Run Forward End

Example mutation

A screenshot of the Avida-ED web application interface, showing a simulation of a population of organisms in a dish. The interface includes a top navigation bar, a left sidebar with configuration options, a central simulation window, and a right sidebar with environmental settings.

Top Navigation Bar: Contains browser tabs (ML, Copy, ALife, Count, SLIDE, Final, Avida, Avid X), a login button, and a list of open files (untitled, Learn, PDF, PDF, Card).

Left Sidebar: Includes buttons for Population, Organism, and Analysis. Below these is the Freezer section, which lists configured dishes (@default, Only Notose, Testing Config), organisms (@all_functions, @ancestor, Does one not), and populated dishes (@example, @popdshexp).

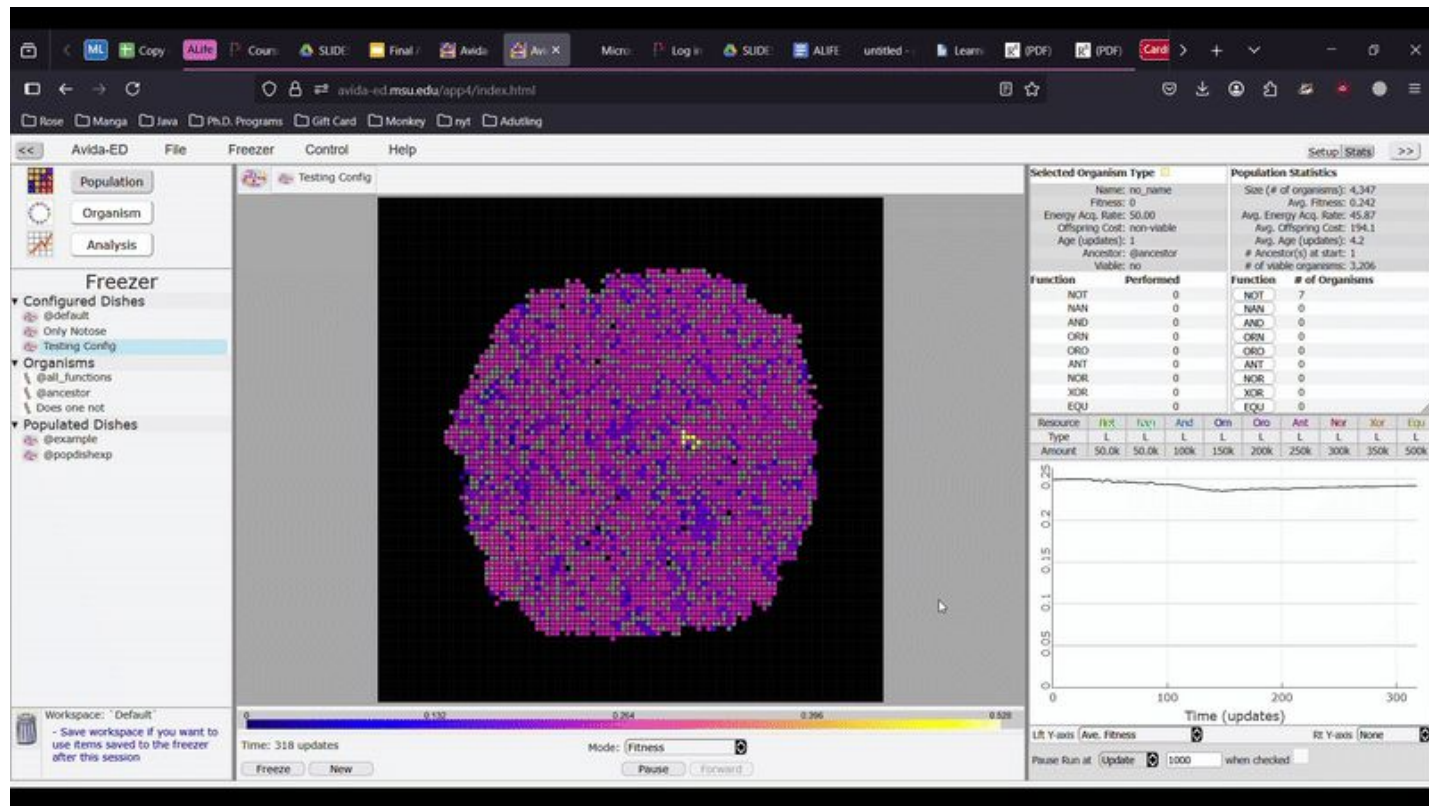
Central Simulation Window: Displays a circular dish filled with a dense population of organisms, represented by small colored dots. The organisms are primarily yellow and orange, with some purple and blue dots scattered throughout.

Right Sidebar: Contains the Environmental Settings section, which includes a Dish Size (100 x 100 cells = 10000 cells), Per Site Mutation Rate (2.0 %), Ancestral Organism(s) (@ancestor), and Place Offspring (Near their parent or Anywhere, randomly). Below this is a table of Resource Options, showing the strength of energy reward for various organisms.

Resource	Limit	Initial / cell	Strength
Notose	Limited	5	Easy (x2)
Flanose	Limited	5	Easy (x2)
Andose	Limited	10	Moderate (x4)
Onose	Limited	15	Moderate (x4)
Orose	Limited	20	Hard (x8)
Antose	Limited	25	Hard (x8)
Norose	Limited	30	Very Hard (x16)
Xorose	Limited	35	Very Hard (x16)
Equose	Limited	50	Brutal (x32)

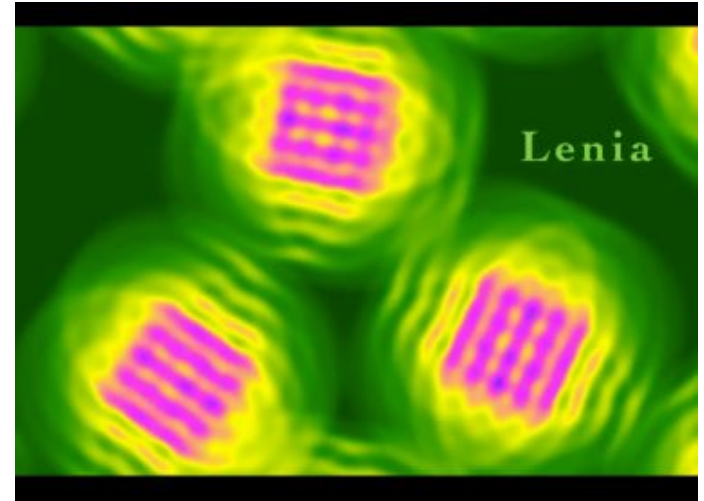
Bottom Bar: Includes a Workspace section (Default) with a note to save workspace to the freezer, a Time display (279 updates), and a Mode selector (Fitness).

Example mutation



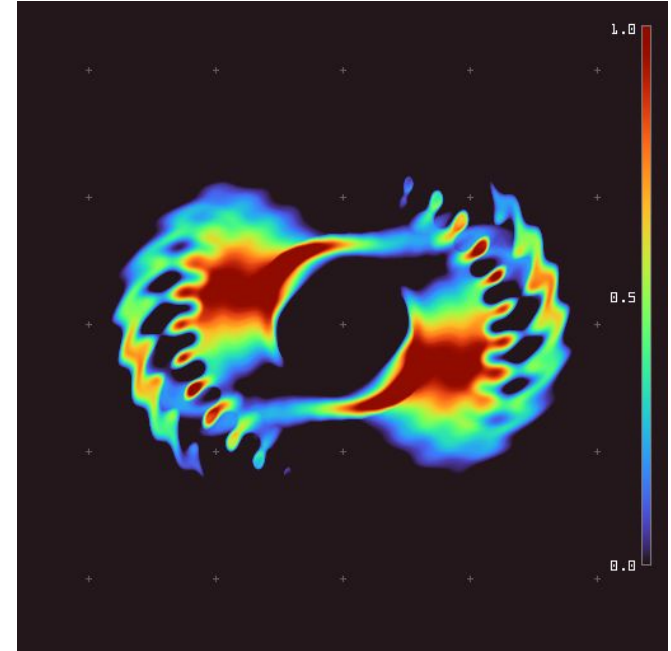
Will Greenlee - Lenia

- Created in 2018-19 by Bert Wang-Chak Chan
 - Name from Latin lenis: “smooth”
 - “Computer simulations show that Lenia supports a great diversity of complex autonomous patterns or “life forms” bearing resemblance to real-world microscopic organisms”
 - Written in three major languages
 - Python
 - Matlab
 - Javascript (browser!)
 - <https://chakazul.github.io/Lenia/JavaScript/Lenia.html>



Will Greenlee - Lenia

- Cellular Automata
 - Generalization of Conway's Game of Life
 - Continuous states, space, and time
- “Lenia” (name of the organisms) are
 - Geometric
 - radial, bilateral, or rotational symmetry
 - Metametric
 - modular repetition of body parts or structures
 - Fuzzy
 - soft boundaries and values, not discrete
 - Resilient
 - retain their properties over time
 - Adaptive
 - change behavior in response to stimuli





Polymorphic Cells



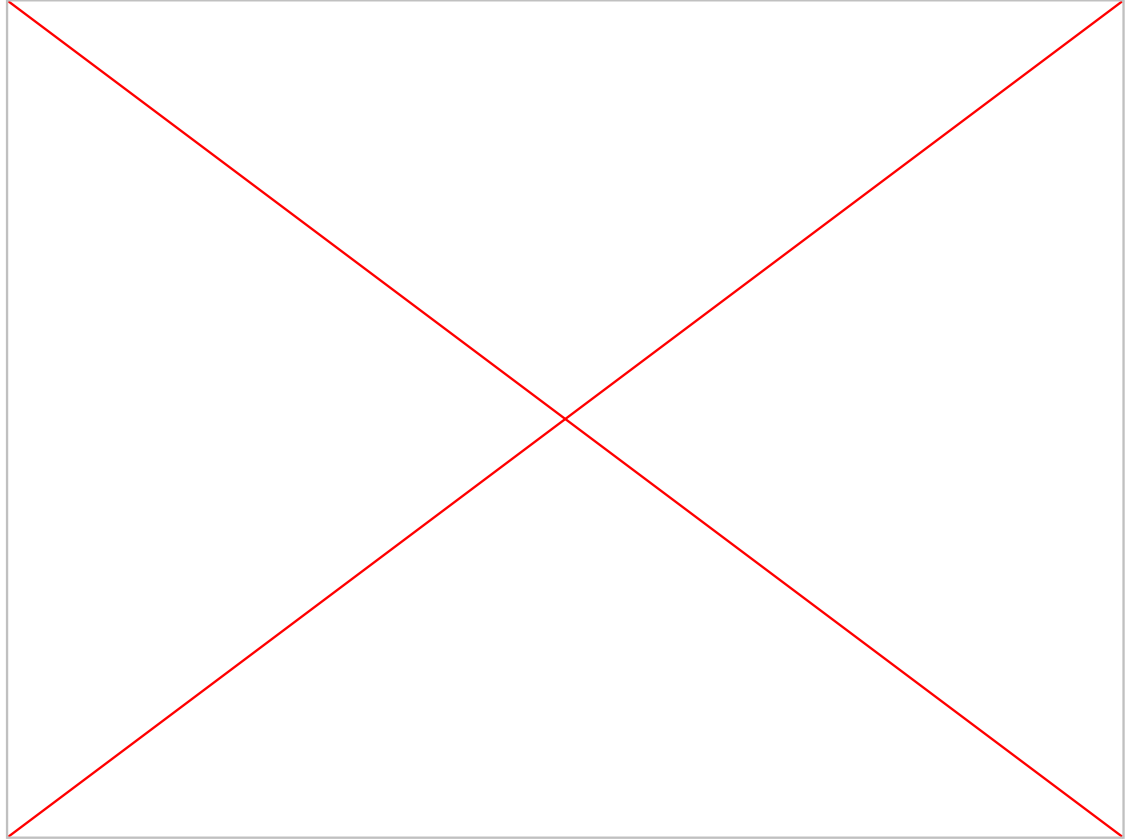
Will Greenlee - Lenia

- Inspirations
 - Conway's Game of Life
 - Reaction-diffusion systems (e.g., Turing patterns).
 - The idea of digital creatures or proto-organisms that display autonomy, stability, and motility.
- Lenia is a superset (sort of)
 - Can approximate Conway's Game of Life
 - $R = 2$
 - $\mu = 0.35$
 - $\sigma = 0.07$
 - Smallest possible kernel

Still lifes		Oscillators		Spaceships	
Block		Blinker (period 2)		Glider	
Beehive		Toad (period 2)		Light-weight spaceship (LWSS)	
Loaf		Beacon (period 2)		Middle-weight spaceship (MWSS)	
Boat		Pulsar (period 3)		Heavy-weight spaceship (HWSS)	
Tub		Pentadecathlon (period 15)			

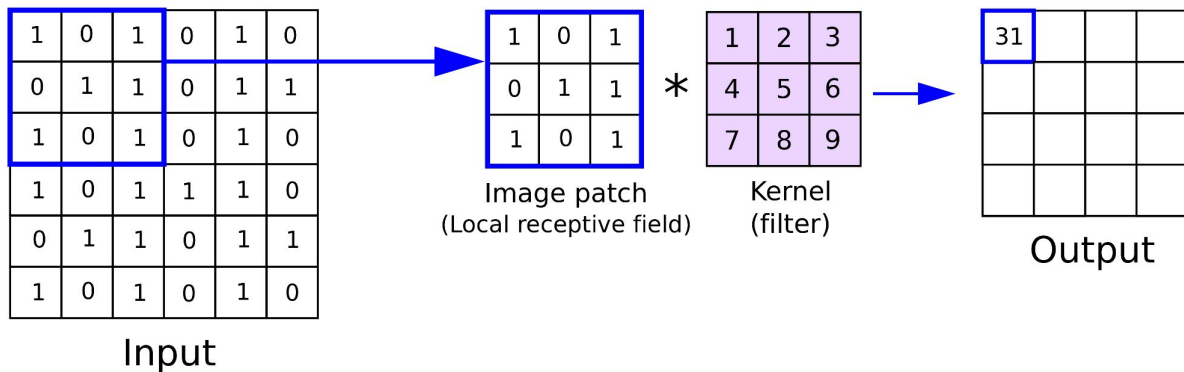
Will Greenlee - Lenia

- Experiment 1



- Lenia operates on:

- A continuous grid
 - $A \subset \mathbb{R}^n$
 - n is the dimension of the simulation, for this set of experiments it is 2
- Real-valued state at each grid point
 - $u(x,t) \in [0,1]$
 - rendered as a heat-map of pixels
- Time evolution is governed by three processes
 - Kernel Convolution, growth mapping, and state update rules (just like discrete CA)



Will Greenlee - Lenia

- Kernel convolution
 - The current state μ is convolved with a kernel K to compute the potential field U
- Growth mapping
 - μ = growth center, σ = growth width.
- Update rule
 - Δt : time step.
 - Clip keeps values in limits of $u(x,t) \in [0,1]$

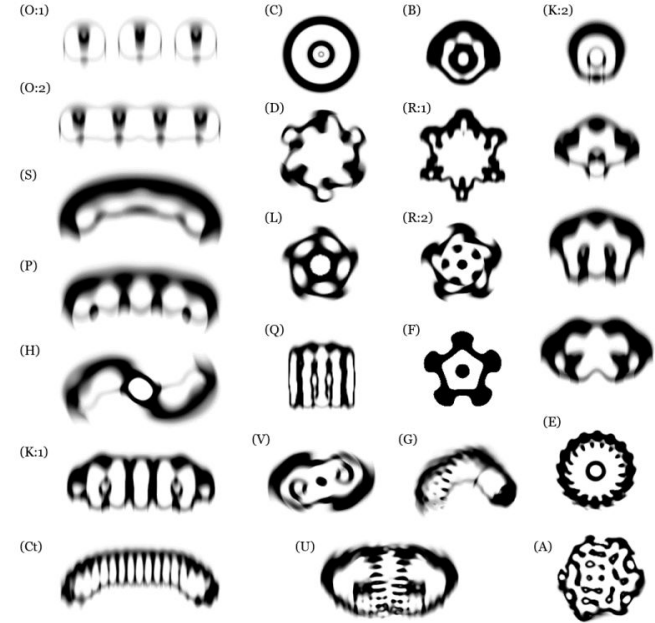
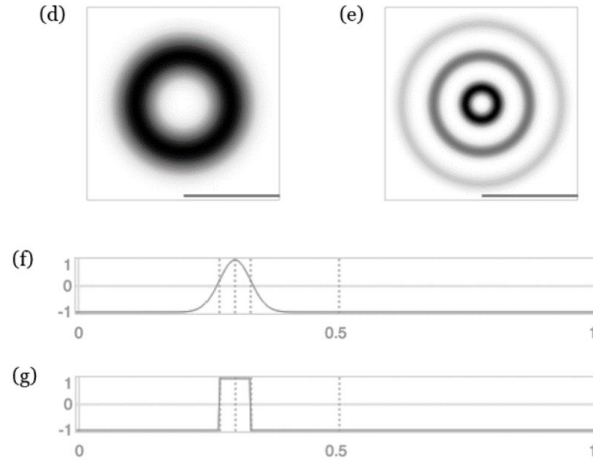
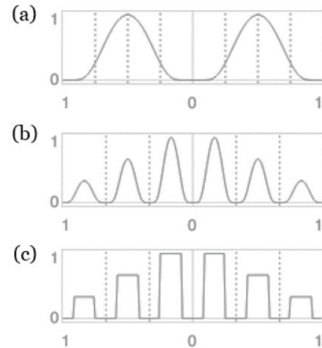
$$U = K * u$$

$$G(U) = 2 \cdot \exp\left(-\frac{(U - \mu)^2}{2\sigma^2}\right) - 1$$

$$u(t+1) = \text{clip}(u(t) + \Delta t \cdot G(U), 0, 1)$$

Will Greenlee - Lenia

- Kernel Types
 - Exponential decay
 - Bell curve (Gaussian)
 - Multi-peak
- Kernel is nearly analogous to species



Will Greenlee - Lenia

- Key parameters:
 - μ : Growth center
 - σ : Growth width
 - R : Kernel radius
 - dt : Time resolution



Params: $\mu = 0.47$ $R = 26$ cells per kernel radius Field: cells, each pixels

$\sigma = 0.105$ $T = 10$ steps per unit time CPU: fps

± 0.001 $\beta = -$ $\Delta t = 0.100 \mu s$

$\eta = -$ $t = 48.200 \mu s$

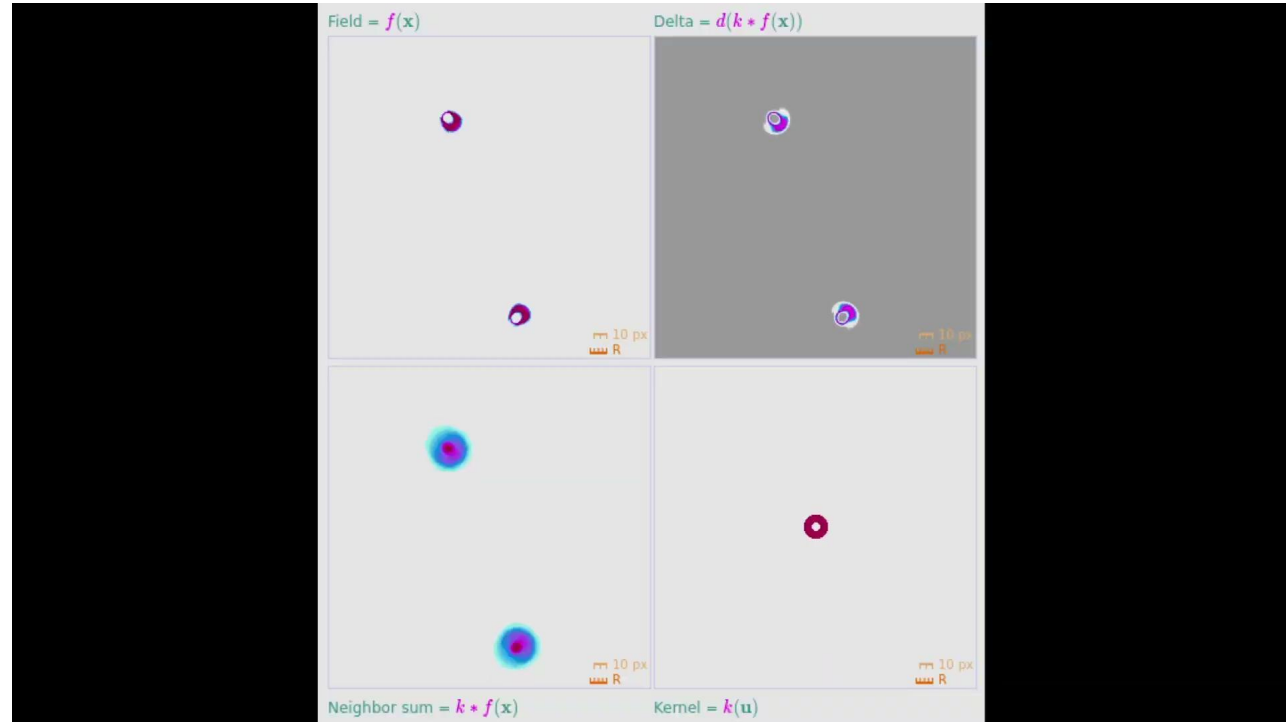
Calc stats: Period: μs

Mass (log scale)

$m = \sum f(x)/R^2 \quad [\mu g]$

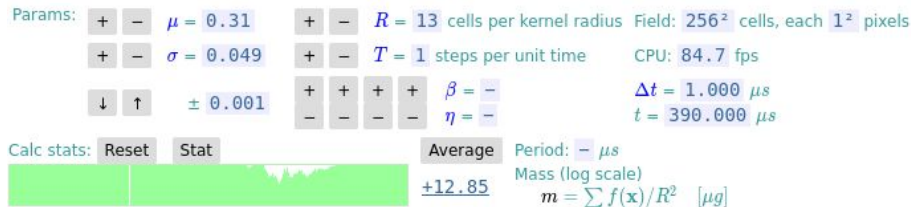
Will Greenlee - Lenia

- Experiment 2



Will Greenlee - Lenia

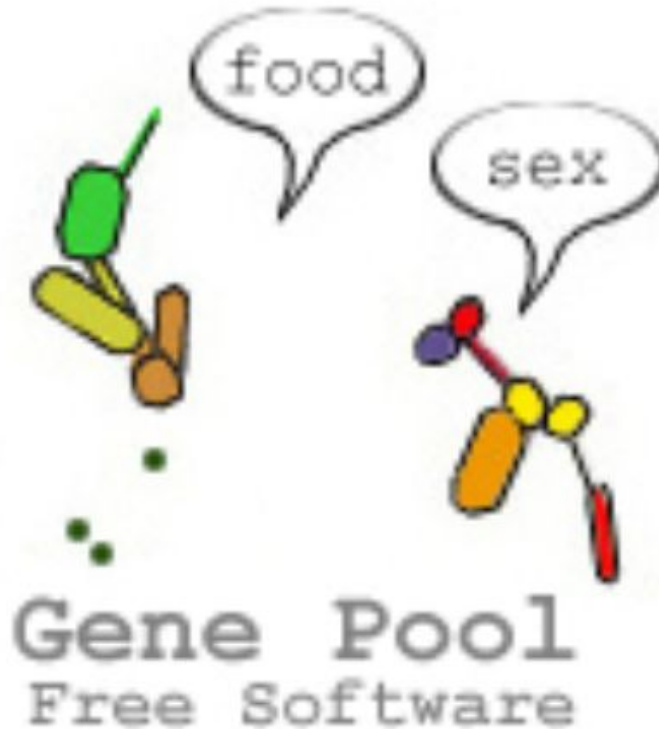
kernel



- Is it repeatable?
 - **Yes! (seeded randomness)**
- Key configuration features
 - The kernel and the growth parameters (μ = growth center, σ = growth width)
- Similar to the Edge of Chaos we discussed early in the quarter!
 - From randomness (the fat glider chaos interactions) | High entropy
 - Comes order (the regular gliders) | Low entropy
- DST
 - The normal gliders are attractors
 - The system tends toward this state
 - The fat gliders are semi-stable, and produce stable points chaotically
- Can be seen almost like a brain
 - Gliders are signals
 - Fat gliders are signal sources (not quite neurons though)
- Future Ideas
 - (If I could get the python running) design the fat gliders from the start
 - Try to create a more stable instance of the fat glider



Steven Johnson - GenePool



icon taken from the creator of
the simulator's website:
<https://www.ventrella.com/Alife/alife.html>

Steven Johnson - GenePool

jeffrey
ventrella



[CV](#)

ventrella.com

[wikipedia](#)

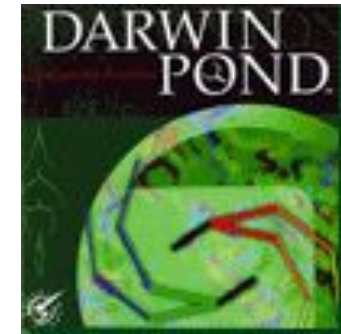
[videos](#)

I am an algorithmic artist, animated artificial life developer, interaction designer, virtual world builder, and visual language practitioner, working at the intersection of art, science, and math. I am antidisiplinarian by nature.

Steven Johnson - GenePool

History

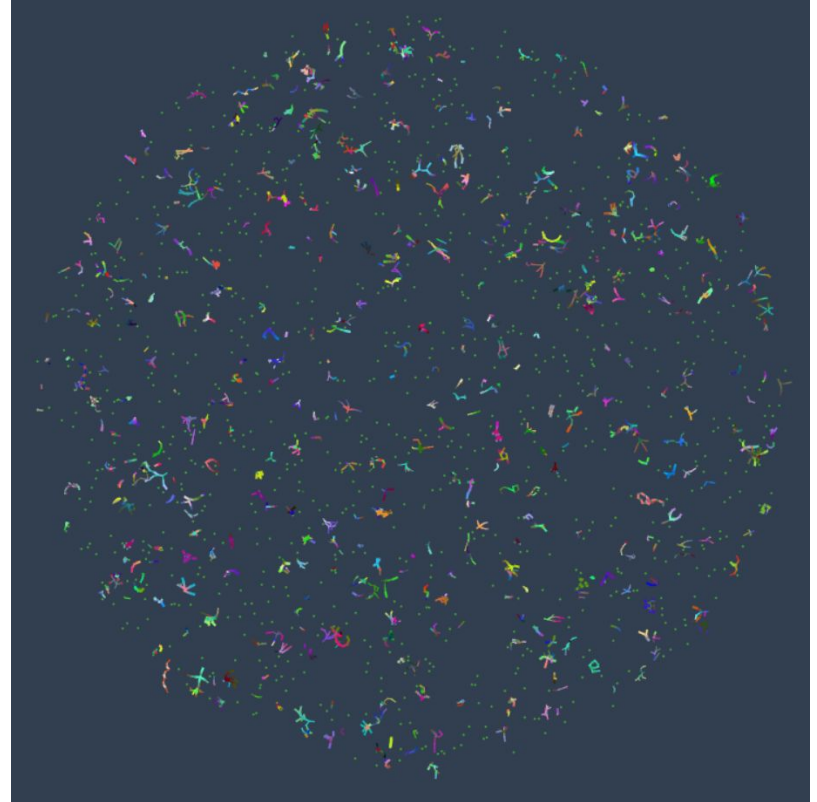
- 1996 - Darwin Pond
- “Darwin Pond is for nature lovers, who spend countless hours (or a lifetime) meditating on the wonders and intricacies of life unfolding.”
- “In Darwin Pond, hundreds of physically-based organisms achieve locomotion via genetically-based motor control and morphology. The ability to have more offspring is a direct outcome of two factors: 1) better ability to swim to within a critical distance to a chosen mate, and 2), the ability to attract other organisms who want to mate.”
- Rocket Science Games -> research project focus ('96) -> game focus ('97) -> studio goes down project, remains free for all users ('98)



Steven Johnson - GenePool

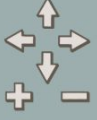
History

- 1998 - GenePool
 - Less interactive features (can't widget your own worms anymore)
 - Added more attractors (sex preferences) to experiment with
 - Ported to local client, website, and mobile eventually
 - Continually added upon and experimented with ever since
- 2021 - newest update (what I ran on)



Steven Johnson - GenePool: Settings

Gene Pool swimbots.com



whole pool

selected

most prolific

oldest virgin

autotracking

mutual love

most efficient

glutton

mouth/genital

rendering OFF

fastest

freeze

pool

swimbot

tweak

graph

info

1. This is Gene Pool

Witness the effects of Darwinian evolution as hundreds of simulated organisms compete for mates and food. Explore this virtual aquarium of proto-swimming robots ('swimbots'), and see how mate preference can affect the course of evolution.

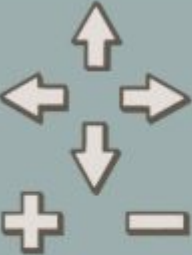
Watch the emergence of a dominant race of swimmers in about 30 minutes. Keep it running for longer and swimming skills will improve, where swimming 'skill' is determined entirely by natural selection.

Every time you start a new pool, the outcome will be different.

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next

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whole pool

selected

most prolific

oldest virgin

autotracking

mutual love

most efficient

glutton

mouth/genital

rendering OFF

fastest

freeze

Viewer Panel

Steven Johnson - GenePool: Settings

pool

swimbot

tweak

graph

info

food growth delay

20

food spread

4000

food bit energy

50

hunger threshold

50

offspring energy ratio

0.5

maximum age

40000

mutation rate

(unknown value)

reset to defaults

Swimbot mate preferences

☐ colorful

☐ not colorful

☒ similar color

☐ big

☐ small

☐ similar size

☐ hyper

☐ still

☐ similar hyper

☐ straight

☐ crooked

☐ similar straight

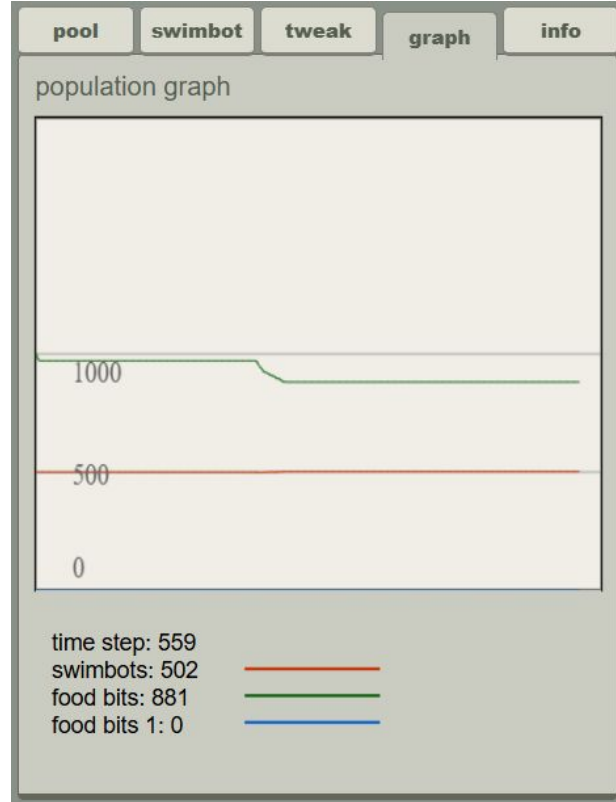
☐ long

☐ short

☐ similar length

☐ random

☐ closest



Steven Johnson - GenePool: Settings

poolswimbottweakgraphinfo

start a new pool with one of these presets...

random	neighborhoods
froggies	**Speciation**
food race	big bang
genetic barrier	empty

SORRY! At the moment, you cannot save or load pool files: the save/load feature of Gene Pool is under construction. I will try to get that done soon. I am switching over to a local file system, using the html file api.

-Jeffrey

After starting a pool from the **Speciation** preset and running it for tens of thousands of time steps, select the button below to get data collected during the simulation. You can load it into [GenePool Lab](https://inegoita.shinyapps.io/gene-pool-laboratory/) (<https://inegoita.shinyapps.io/gene-pool-laboratory/>) (Contact jeffrey@ventrella.com for details).

print swimbot data

poolswimbottweakgraphinfo

No swimbot is selected.
Please select one in the pool. Or...

Create a new swimbot with randomized genes:

create random

Sorry, but currently you cannot load swimbot genes. The code for that is being rewritten.

Create a new swimbot with genes from one of these presets:

Darwin	Margulis
Wallace	Wilson
Mendel	Dawkins
Turing	Dennett

poolswimbottweakgraphinfo

1. This is Gene Pool

Witness the effects of Darwinian evolution as hundreds of simulated organisms compete for mates and food. Explore this virtual aquarium of proto-swimming robots ('swimbots'), and see how mate preference can affect the course of evolution.

Watch the emergence of a dominant race of swimmers in about 30 minutes. Keep it running for longer and swimming skills will improve, where swimming 'skill' is determined entirely by natural selection.

Every time you start a new pool, the outcome will be different.

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next

Steven Johnson - GenePool

Mechanisms/Flow

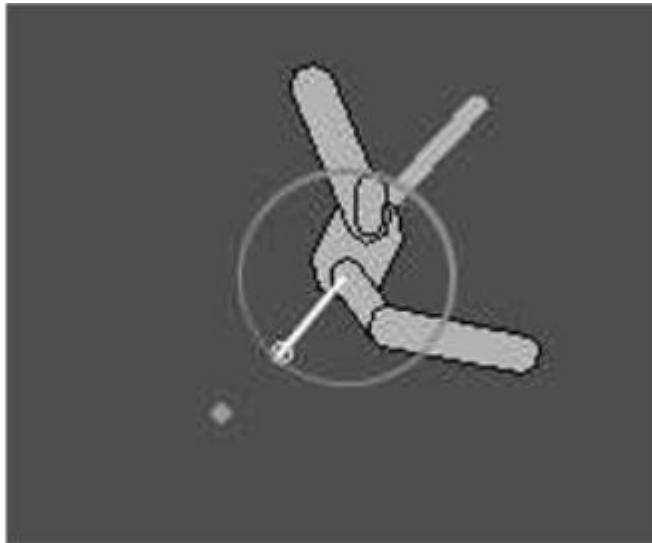


Figure 1.5. The mouth vector and a circle showing the critical distance for eating a food bit.

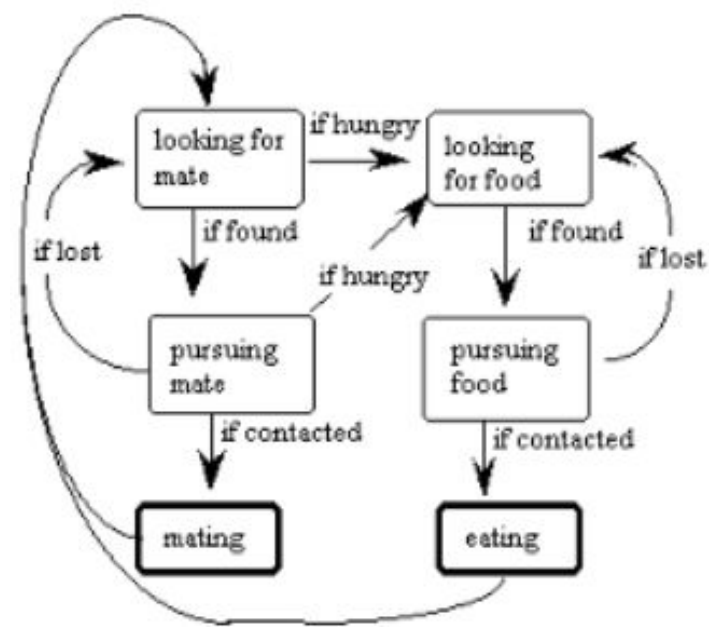


Figure 1.6. Swimbot mental states

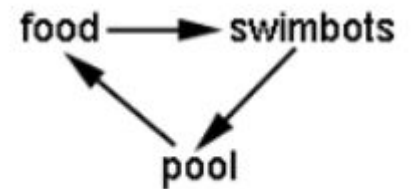


Figure 1.7. Energy flow in GenePool.

Steven Johnson - GenePool

Mechanisms/Flow

Innate orientation based on axis of main body part.

When Goal Present:

- Direction from swimbot to goal is compared to its orientation repeatedly
- Size and sign of angle used to modify all part motions

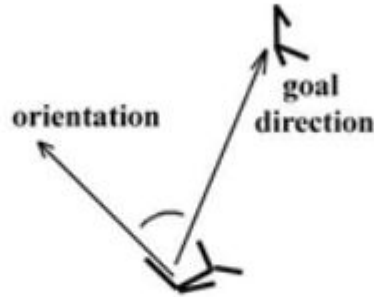


Figure 1.8. Swimbot orientation compared to goal direction modifies genetically-determined turning mechanisms.

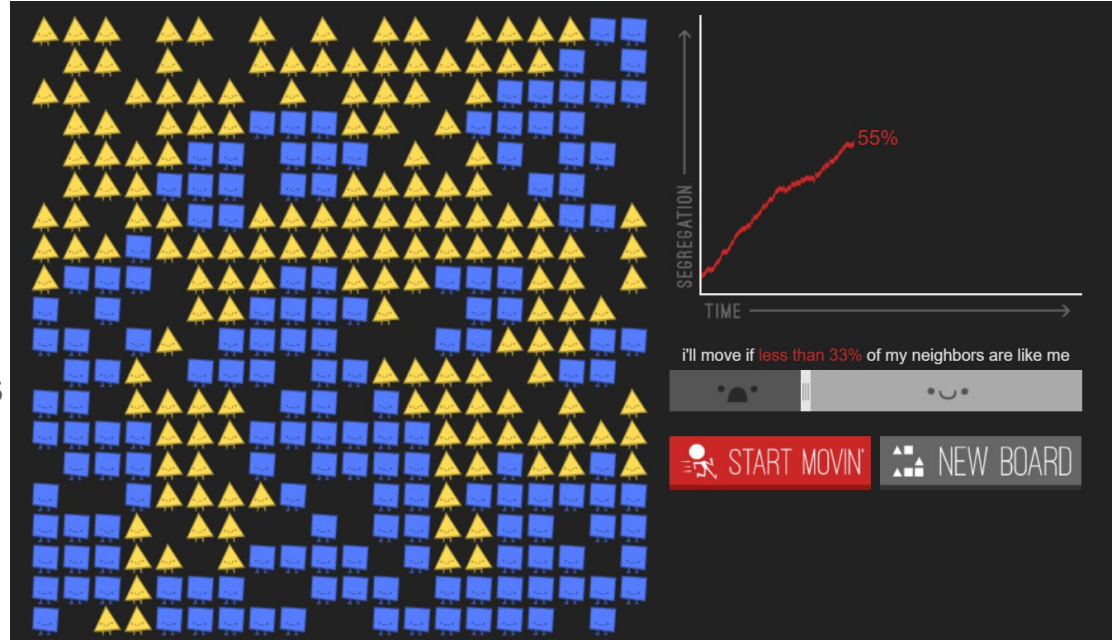
The Questions

Main Questions Answered

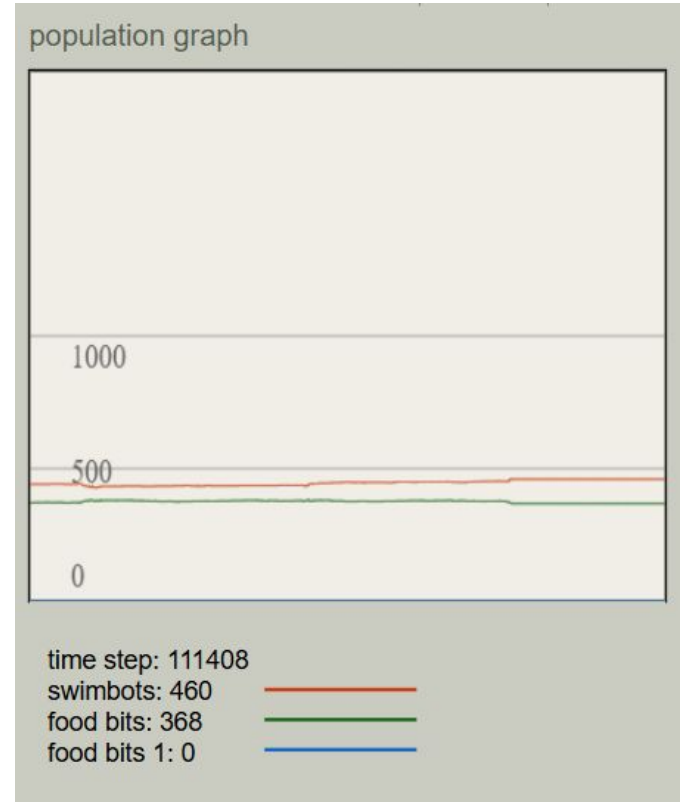
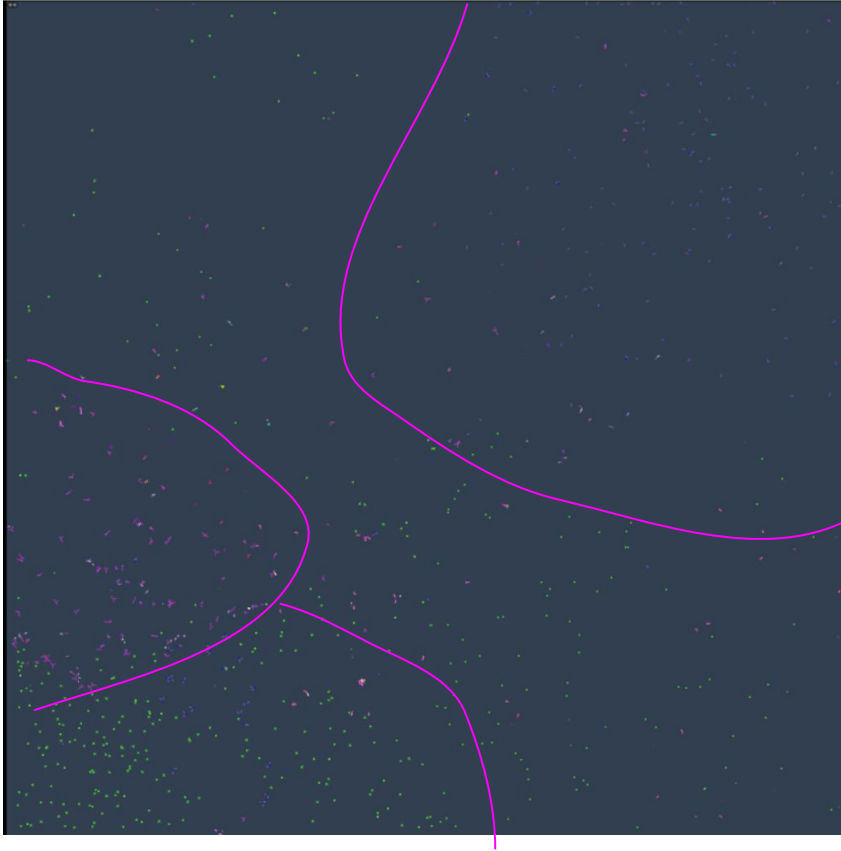
- What is the simulator focused on?
 - Exploring evolution in real-time in an easy-to-access way allowing the user to come up and test their own hypotheses.
- What levels does it attempt to capture?
 - Locomotion, aesthetics, space, food, and mating.
 - “bring some basic principles of evolution to light”
- What is an example of a research question it purports to be able to help answer?
 - Using the “still” attractor. Hypothesis: They go extinct.
 - What happened: sexual dimorphism
 - Majority: sexy “sitters” (they don’t move)
 - Minority: agile “breeders” (they have tail-whips allowing them to move around really fast)

My Experiment

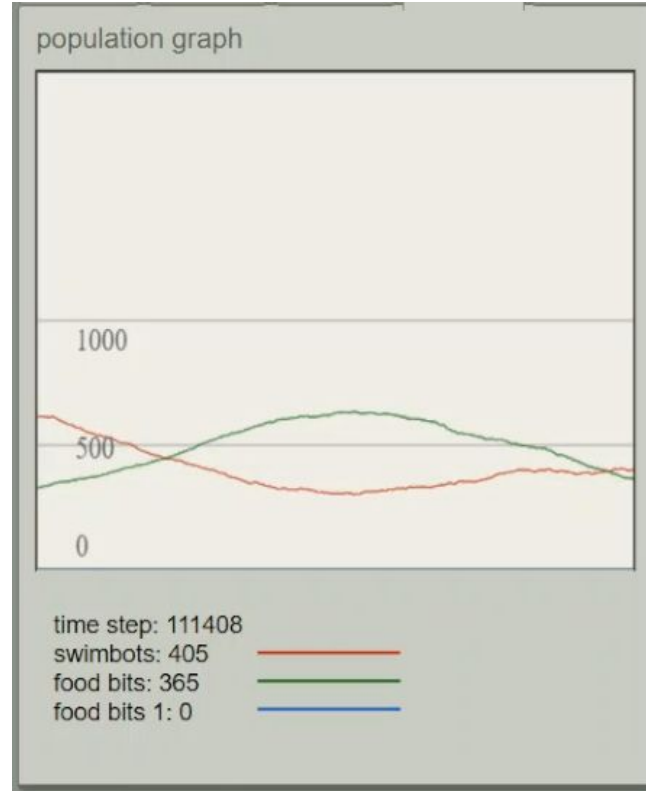
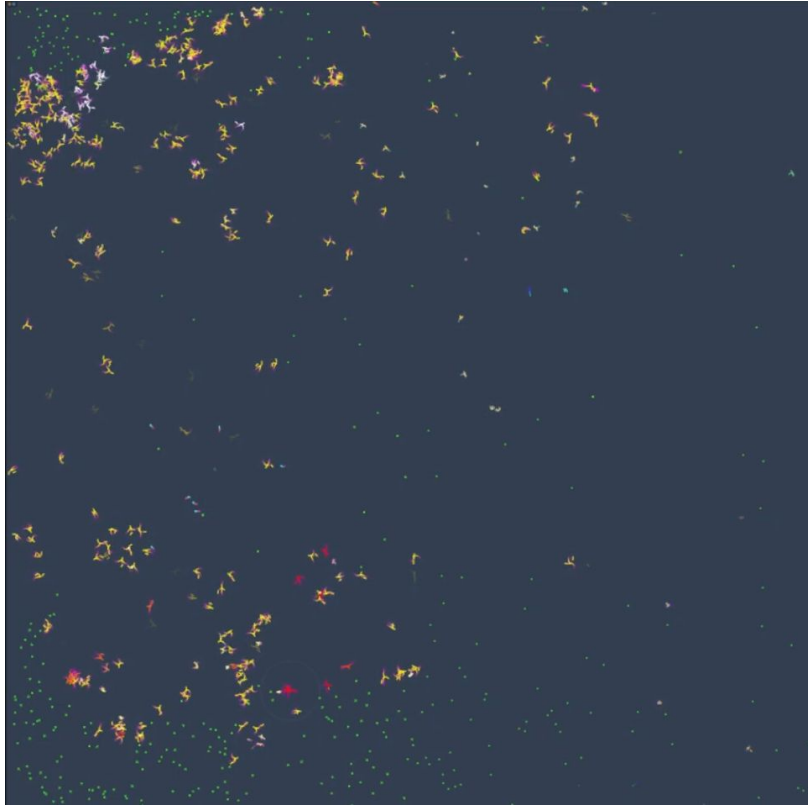
- Artificial Society inspiration
 - Polygon Segregation
- Question
 - Will the more advanced GenePool result in segregation like polygons did?
- Approach
 - Default settings (including “Similar color” attractor)



My Experiment: Results (Run 1 - no video)



My Experiment: Results (Run 2 - video)



Interesting results:

- Food geographic disparities
- One race emergence
- Mutations (red species) found in video

My Experiment: Required Questions

- Is it repeatable?
 - Yes.
- What are the most important mechanisms (change/remove components)?
 - Attraction rule (outside my experiment - access to food, no food = death)
- Can we understand in detail one specific instance of the observed results/behavior?
 - Similar color attractor leads to segregated populations just like the shapeism rule in polygons
- Could we possibly understand the system better through information theory?
 - Using entropy (randomness, uncertainty) and applying it to diversity. In this case, a higher entropy would mean higher diversity and we could then graph that over time to visualize how diversity evolves throughout the simulation.
- Could we possibly understand the system better through dynamical systems theory?
 - Attractors being a mix between the attractor rules and the proximity to food
- Is there potential of seeing this behavior/phenomena in the real world?
 - How could you try to imitate this?
 - Gene editing
 - What would be crucial to consider in evaluating this?
 - Comparisons to simulations and insights gained from anthropology, archaeology, ecologist, and paleontology.

My Experiment: Cross-Reference and Wishes

Cross-Reference:

- Ventrella paper - Celebrating Diversity Experiment
 - “similar color” criteria
 - “swimbots will choose mates whose bodies contain the closest spectrum of colors to their own”
 - “encourage interracial mating by adding a new attractiveness criterion called “opposite color”. Not surprisingly, when this is turned on, the population converges on a perpetual state of psychedelic diversity.”

Wishes:

- I wanted to run opposite color attractor! (Why isn't it an option!)
- Continual updates after 2021
 - Re-addition of saving and loading in newest update
- A better fast-forward/time slider!! (I had my laptop running the simulator for many hours and if I pulled up something else in full-screen it would stop!!)

Other Things I Did

- Random attractor
- Closest attractor
 - Observation: Took more than double the time as the others to get to similar timesteps
- Recorded videos for all three attractors I tested (hours and hours)
 - Could pull those up and look through them for added insight from the class! :)
- Kept the tabs open
 - If they haven't closed by the time we get into class we can continue running them and see what happens
 - Also, can print out the genetic data of each and compare it to the genetic data of a fresh start

Appendix

Simulation Summary:

5910

Total Swimbots

111363

Total Timesteps

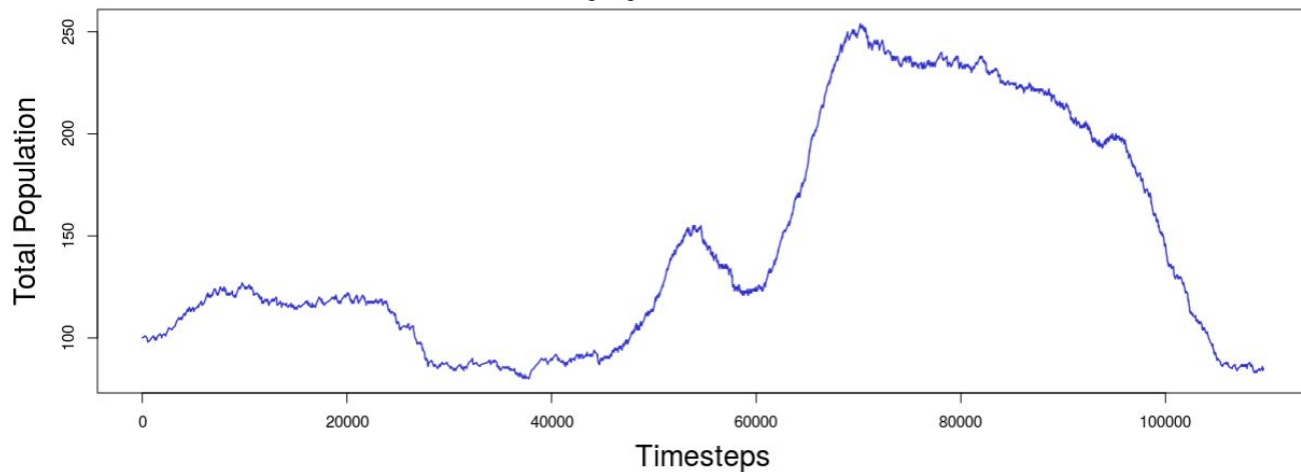
40001

Maximum Lifespan

13598

Average Lifespan

Total population over time



Similar color (R1)

Appendix

Simulation Summary:

6745

Total Swimbots

174878

Total Timesteps

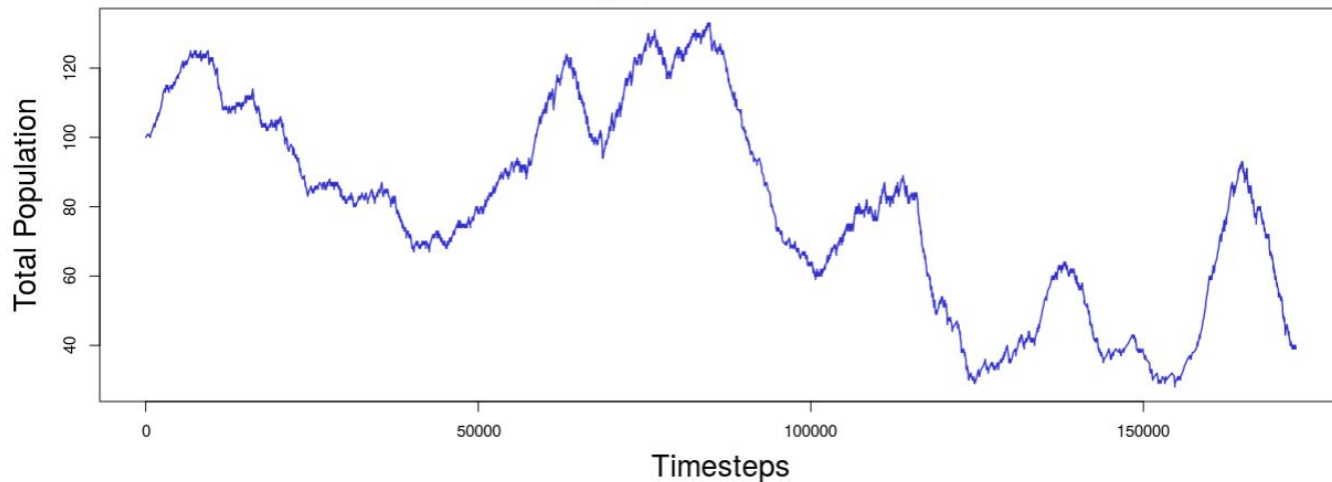
40001

Maximum Lifespan

10358

Average Lifespan

Total population over time



Similar color (R2)

Appendix

Simulation Summary:

4680

Total Swimbots

133296

Total Timesteps

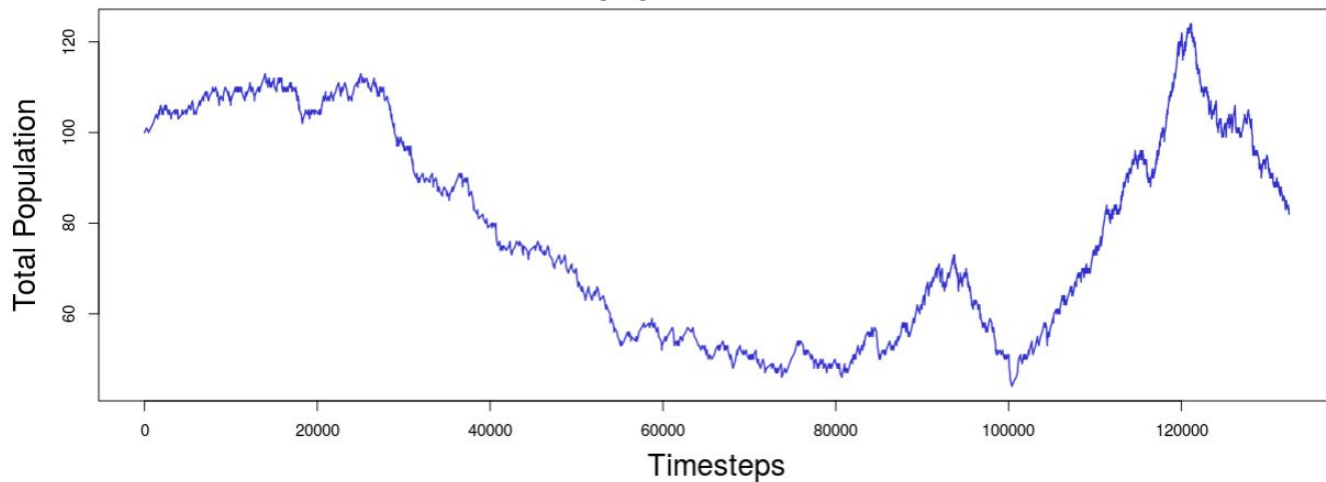
40001

Maximum Lifespan

11144

Average Lifespan

Total population over time



Random

Appendix

Simulation Summary:

8975

Total Swimbots

150788

Total Timesteps

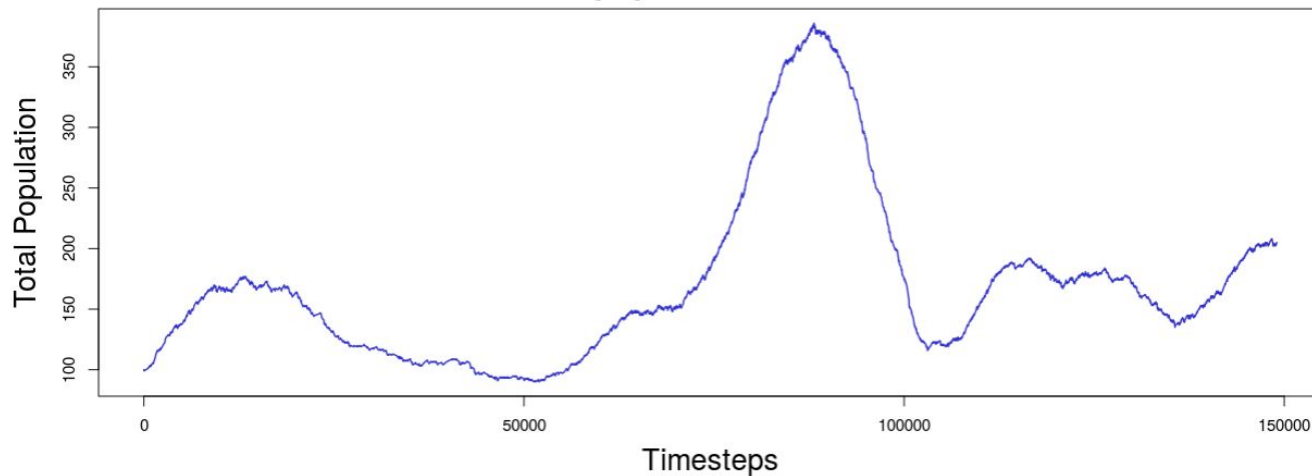
40001

Maximum Lifespan

14269

Average Lifespan

Total population over time



Closest